Introduction to Machine Learning



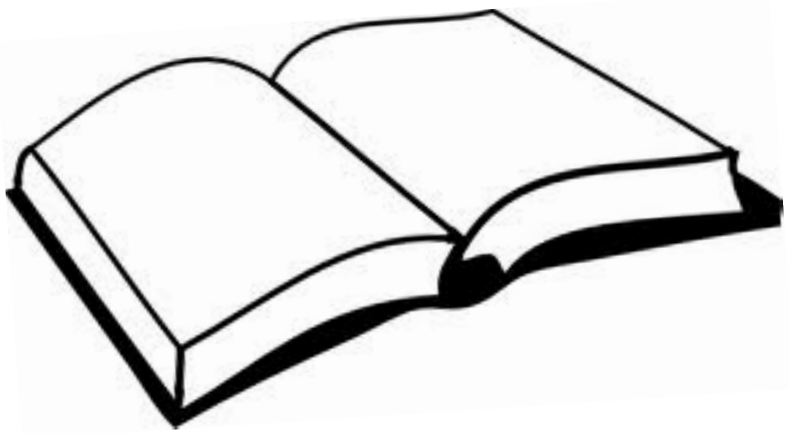
Instructor Name: Asma Khalid Butt

Department of Computer Science, COMSATS University Islamabad - Islamabad Campus - Pakistan

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- Traditional Programming vs Machine Learning
- When is Machine Learning Useful?
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- Types of Machine Learning



What is Machine Learning?

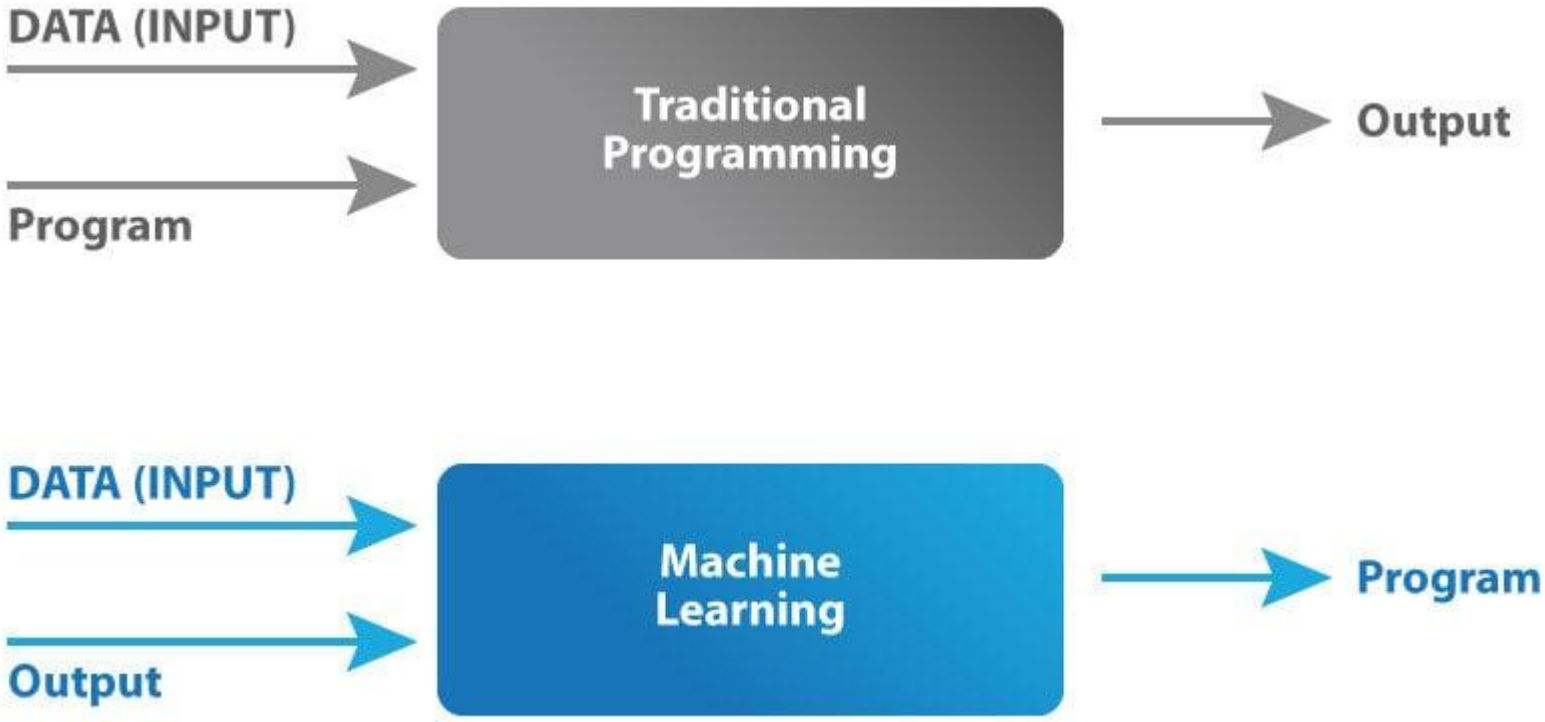
Formal Definition:

The subfield of computer science that "gives computers the ability to learn without being explicitly programmed". (Arthur Samuel, 1959)

In Simple Words:

Machine Learning is about learning from data.

Traditional Programming vs Machine Learning



Example: Addition

Traditional Programming

A program is explicitly written to add two numbers.

Input: 5, 10

Program: Addition logic

Output: 15

If the program is designed only for two numbers, additional cases may require modifying the code.

Machine Learning

Provide examples of numbers and their sums.

Data → Training → Model learns the pattern → Prediction

The model learns the relationship from the examples.

When is Machine Learning Useful?

Machine Learning is particularly useful when:

1. It is difficult to write rules for every case
2. The number of possible cases is extremely large

Use Case 1: Spam Classification

Task:

Determine whether an email is ***Spam or Not Spam***

Traditional Approach:

Create many **if-else rules** based on:

- Keywords
- Frequency of words
- Images
- Other characteristics

Problem in Traditional Approach:

Spammers can change their words and patterns.

Result: Rules need frequent modification.

Spam Classification with ML

Machine Learning Approach

Email Data → ML Algorithm → Spam Classifier

The model learns patterns from existing emails.

If the Data changes, the model can learn new patterns through retraining.

Key Advantage:
Less dependence on manually written rules

Use Case 2: Image Classification

Task:

*Determine whether an image contains a **dog** or not.*

Challenges:

- Hundreds of breeds
- Different sizes
- Different colors
- Different shapes
- Different appearances

Question:

Can we manually write rules for every possible appearance?

No.

How Humans Learn to Recognize Dogs

Humans learn through examples:

- This is a **dog**
- This is not a dog
- This is a **cat**

Our brains learn patterns from examples.

Machine Learning follows a similar principle:

Examples/Data → Learn Patterns → Recognize New Images

A Brief History of Machine Learning

- Machine Learning concepts have existed for decades
- The underlying **Mathematics and Theory** are not new

- Earlier adoption was limited by:
  - Lack of large datasets
  - Difficulty of Data collection and labelling
  - Limited computing power

Why Machine Learning Grew Rapidly

Three Major Factors

1. More Data

- Internet users generate huge amounts of Data
- Digital Data has grown rapidly

2. Better Hardware

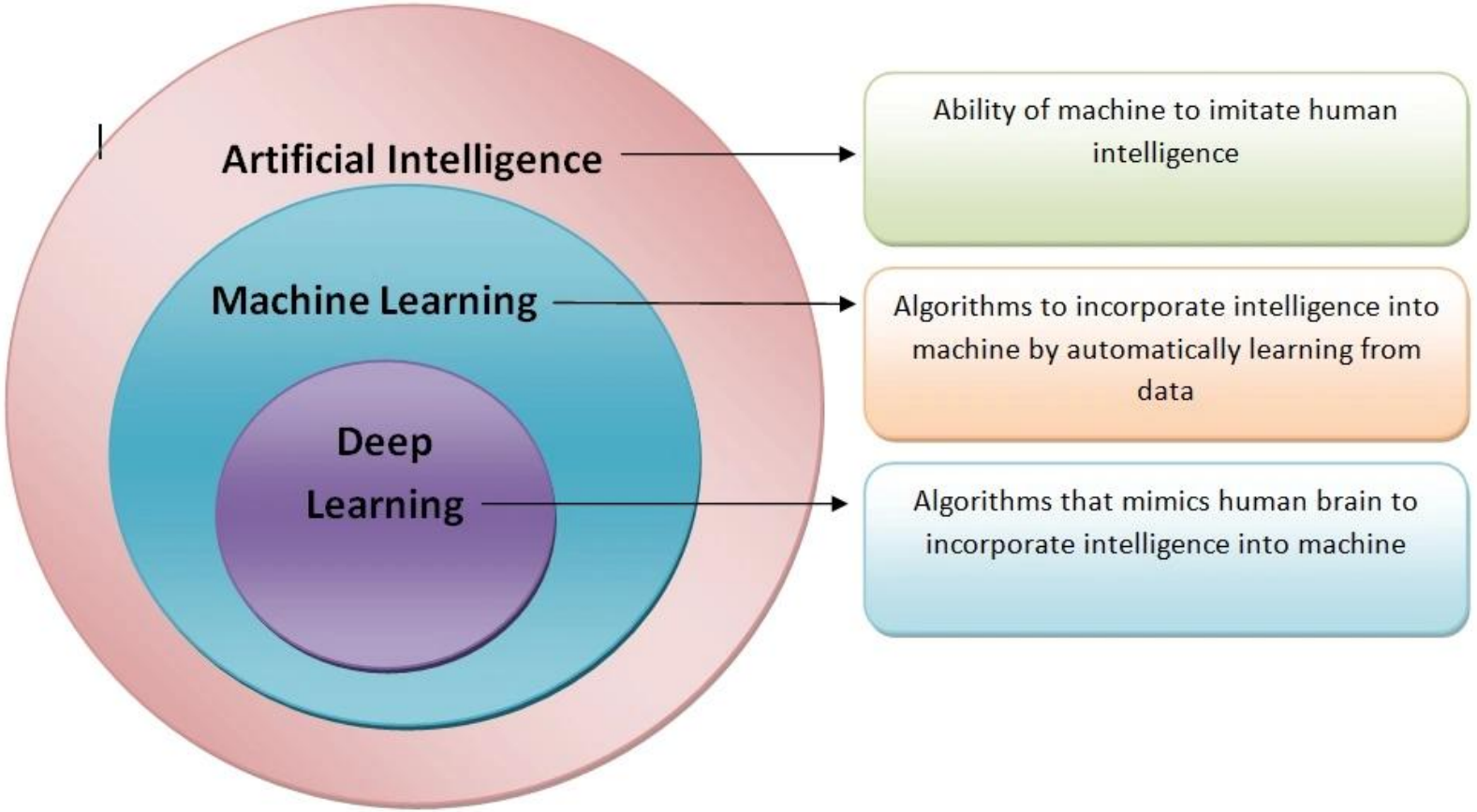
- Powerful CPUs and GPUs
- More RAM
- Faster computation

3. Better Algorithms

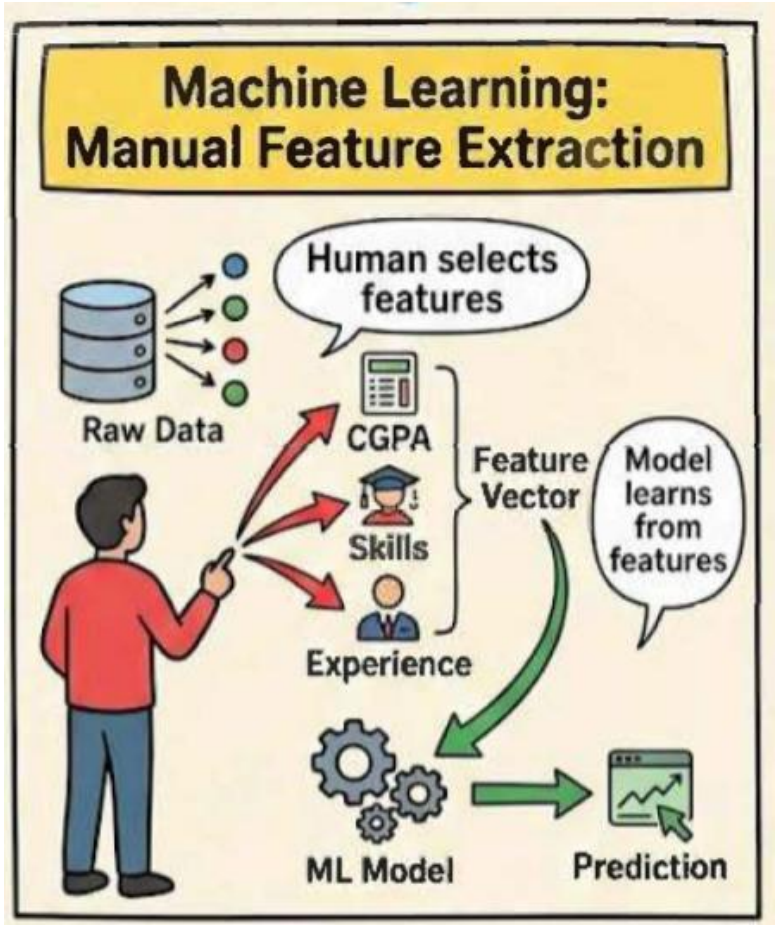
Modern algorithms can utilize large datasets effectively

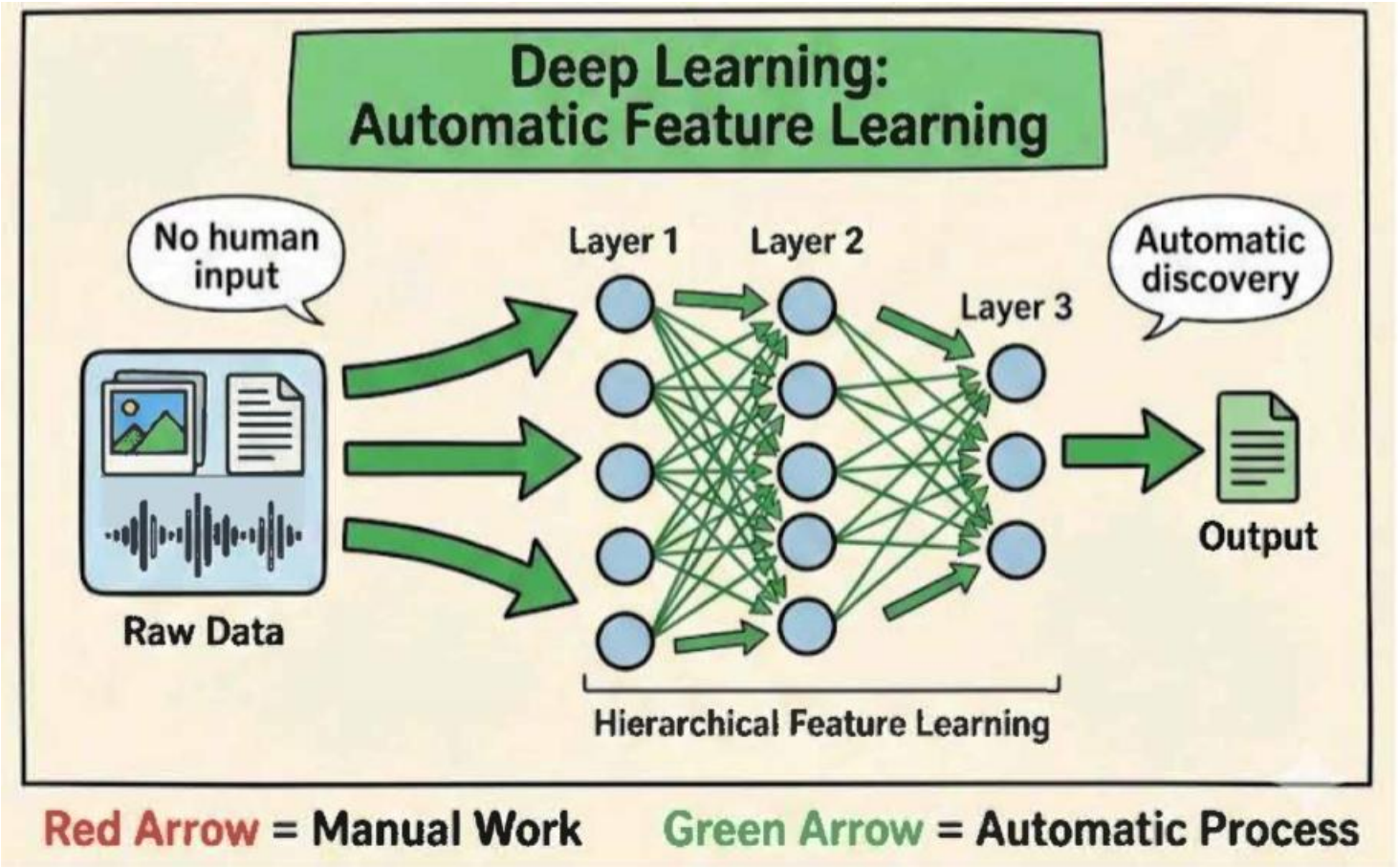
Data + Hardware + Algorithms → ML Growth

AI vs ML vs DL

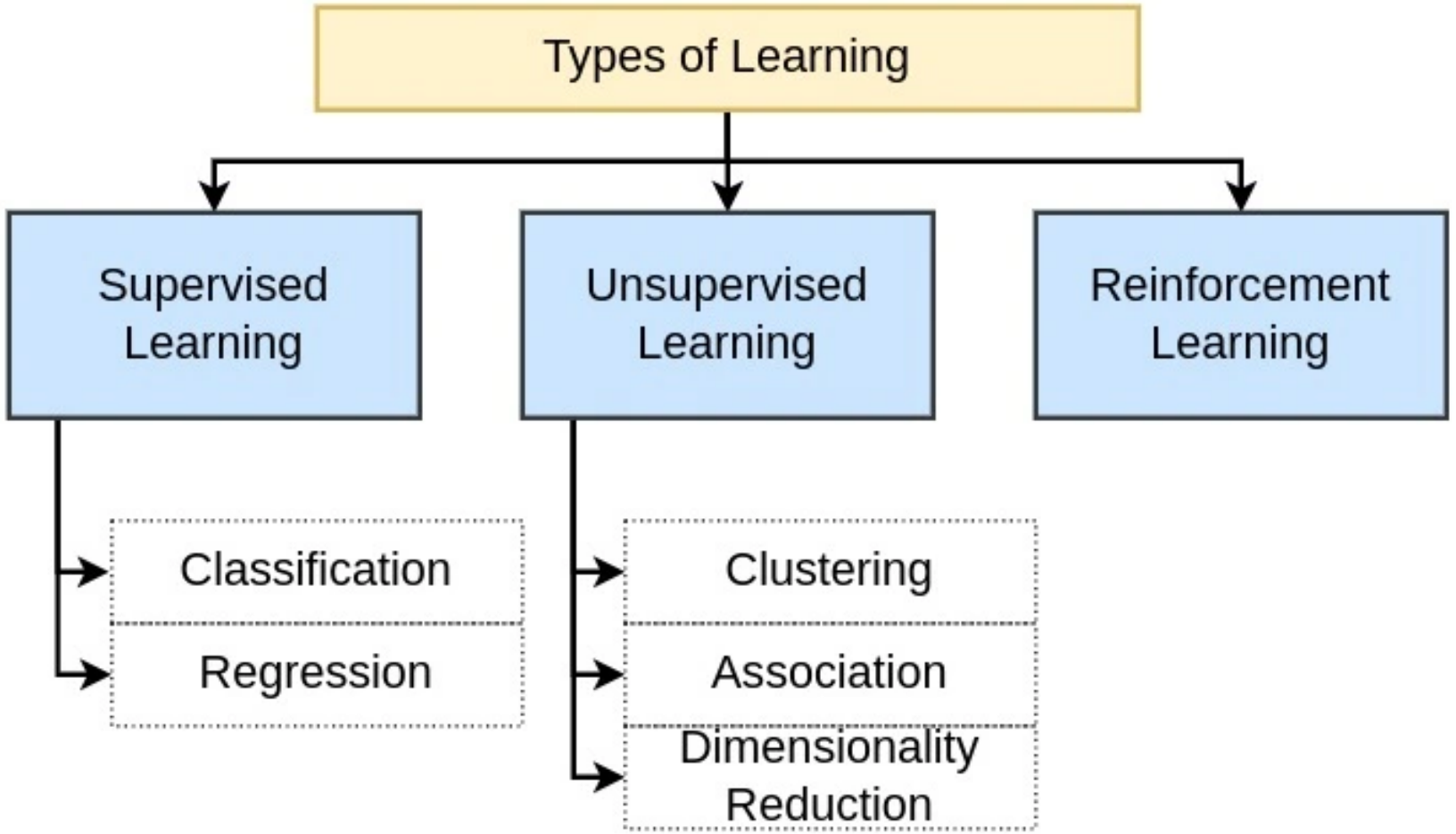

Artificial Intelligence
Machine Learning
Deep Learning
Ability of machine to imitate human intelligence
Algorithms to incorporate intelligence into machine by automatically learning from data
Algorithms that mimics human brain to incorporate intelligence into machine

ML vs DL


Machine Learning:
Manual Feature Extraction
Human selects features
Raw Data
CGPA
Skills
Experience
Feature Vector
Model learns from features
ML Model
Prediction


Deep Learning:
Automatic Feature Learning
No human input
Raw Data
Layer 1
Layer 2
Layer 3
Automatic discovery
Output
Hierarchical Feature Learning
Red Arrow = Manual Work
Green Arrow = Automatic Process

Types of Machine Learning



Supervised Learning

In **Supervised Learning**, the model learns from:

- **Input Data**
- **Output / Target Labels**

The model learns the relationship between **Input → Output**.
Goal: Predict the correct output for new, unseen data.

Example:
Student IQ + CGPA → Placement Prediction

Two Main Types of Supervised Learning

Supervised Learning is mainly divided into:

1. Classification

Output is **Categorical**

Example: Spam / Not Spam

2. Regression

Output is **Numerical**

Example: Predict house price

Classification

Classification is used when the output belongs to a **category/class**.

Examples:

- Spam / Not Spam
- Rain / No Rain
- Dog / No Dog
- Pass / Fail
- Disease / No Disease

Regression

Regression is used when the target/output is a **continuous values**.

Examples:

- House Price Prediction
- Salary Prediction
- Temperature Prediction
- Sales Prediction

Unsupervised Learning

In **Unsupervised Learning**:

- We have **Input Data**
- There are **No Output Labels**

The model discovers hidden patterns or structures in the data.

Goal: Find useful information without predefined labels.

Main Tasks of Unsupervised Learning

Four important tasks:

1. Clustering
2. Dimensionality Reduction
3. Anomaly Detection
4. Association Rule Learning

Clustering

Clustering groups similar data points together.

Example: Students can be grouped based on:

- IQ
- CGPA
- Study habits

Possible groups:

- High-performing students
- Average students
- Low-performing students

No group labels are provided beforehand.

Applications

- Customer segmentation
- Grouping similar news articles
- Organizing photos by similar faces

Dimensionality Reduction

What it does: Reduces the number of features (columns/variables) in your data while keeping the important information.

Why it's needed: Real-world data often has hundreds or thousands of features. This causes problems:

- Hard to visualize (can't plot 100 dimensions)
- Slower computation
- "Curse of dimensionality" — too many features can confuse models

How it works: It finds a smaller set of new features that capture most of the "meaningful variation" in the original data.

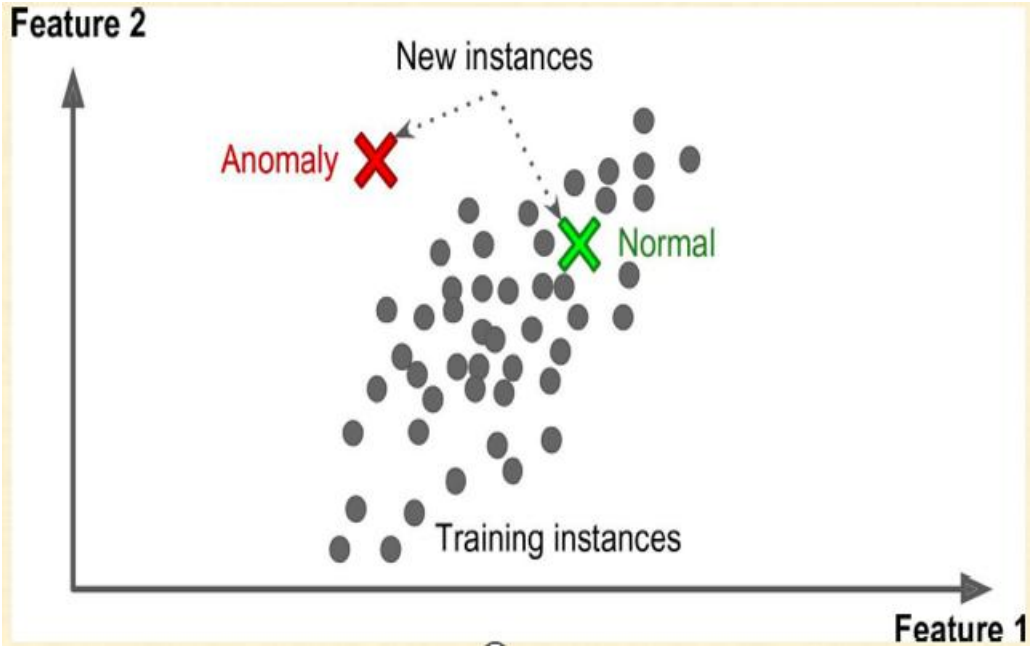
Applications: Data visualization, speeding up other ML models

Anomaly Detection

What it does: Identifies data points that are unusual or don't fit the normal pattern.

How it works: The algorithm learns what "normal" data looks like, then flags anything that deviates significantly from that pattern.

Applications: Unusual transactions, Credit card Fraud detection



Association Rule Learning

What it does: Finds interesting relationships or patterns between variables in large datasets.

How it works: It looks for "if this, then that" patterns — i.e., if event A happens, event B is likely to happen too.

Simple Example (the classic one):
A supermarket analyzes purchase data and finds:

"Customers who buy diapers also tend to buy beer"

This isn't obvious logically, but the data reveals it. Stores can then place these items near each other, or bundle promotions.

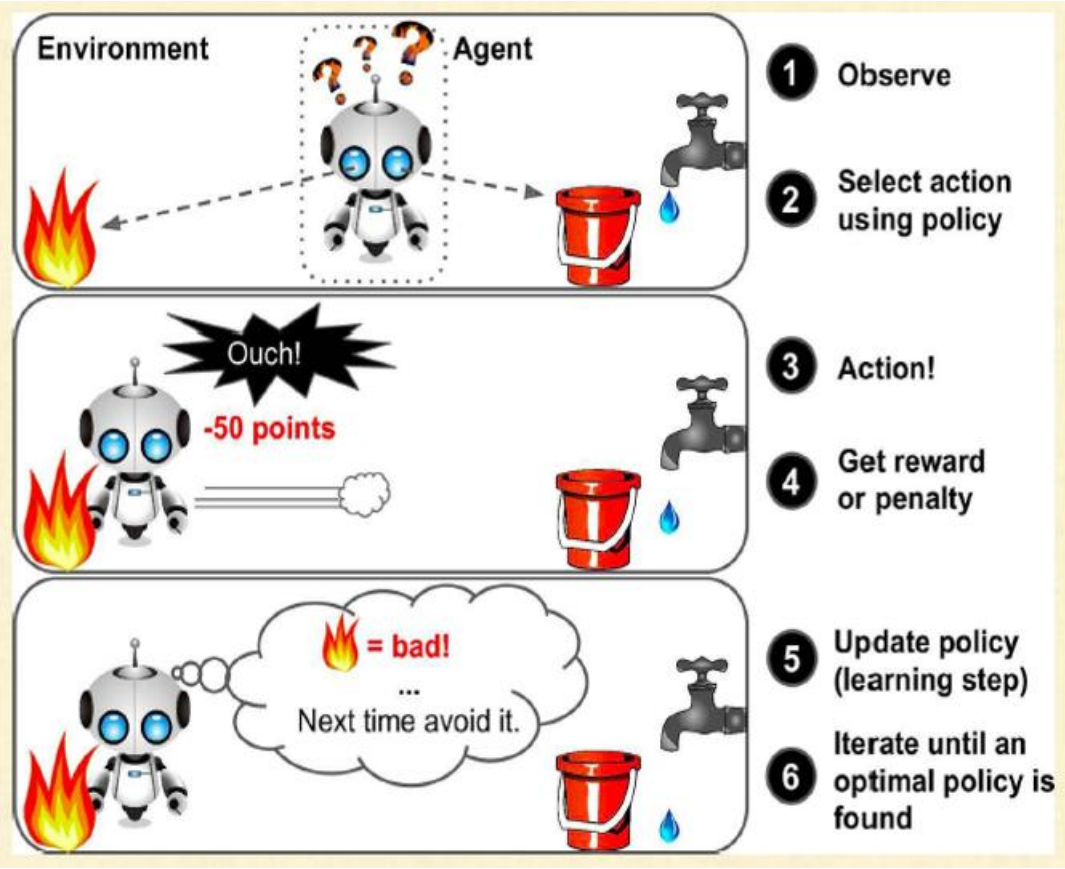
Applications: Market basket analysis (retail), recommendation systems, website navigation pattern analysis

Reinforcement Learning

In **Reinforcement Learning**, an **Agent** learns by interacting with an **Environment**.

The Agent:

- Observes the Environment
- Takes an **Action**
- Receives a **Reward or Penalty**
- Learns which actions are better



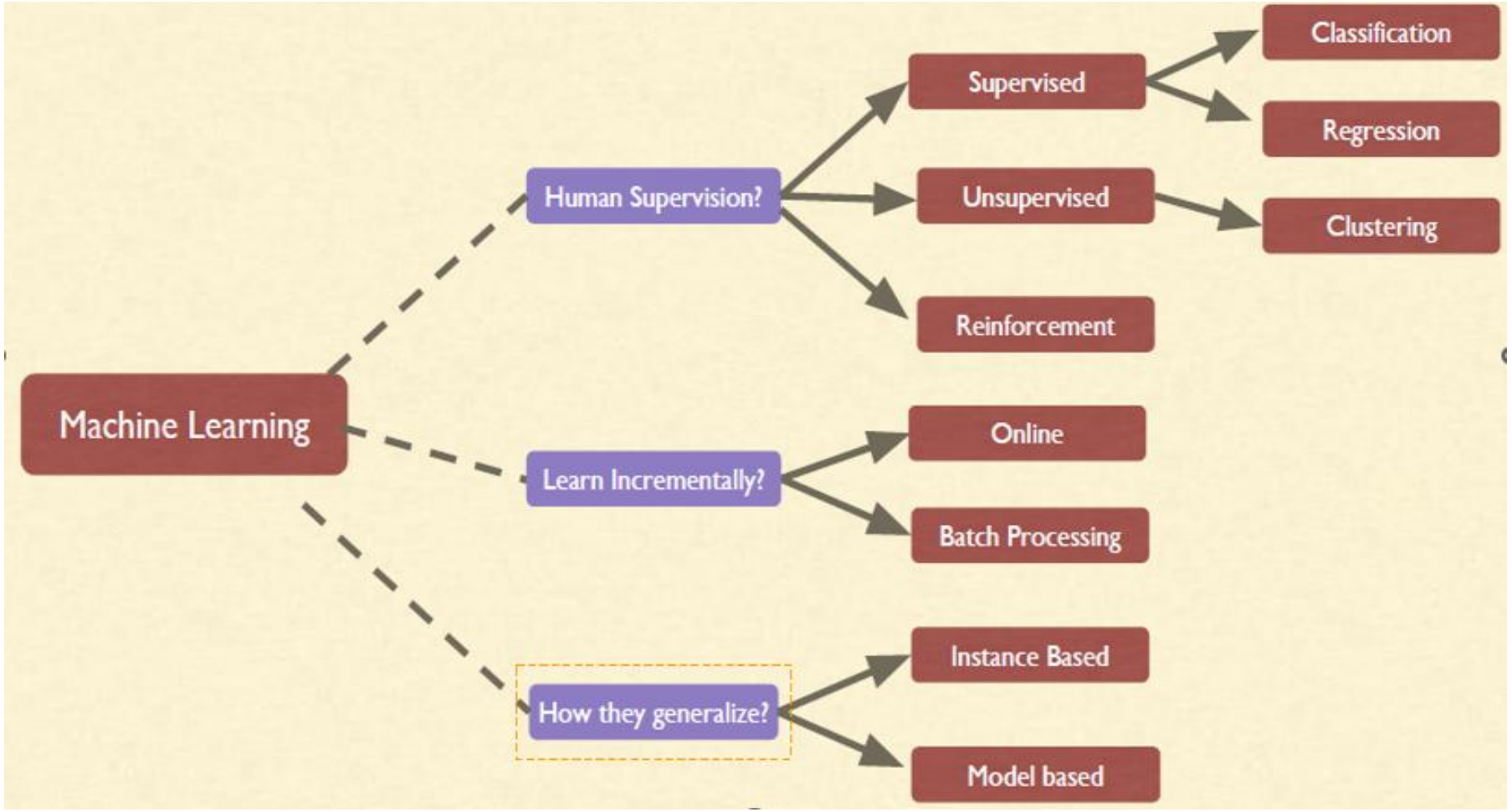
Applications of Reinforcement Learning

Reinforcement Learning is used in:

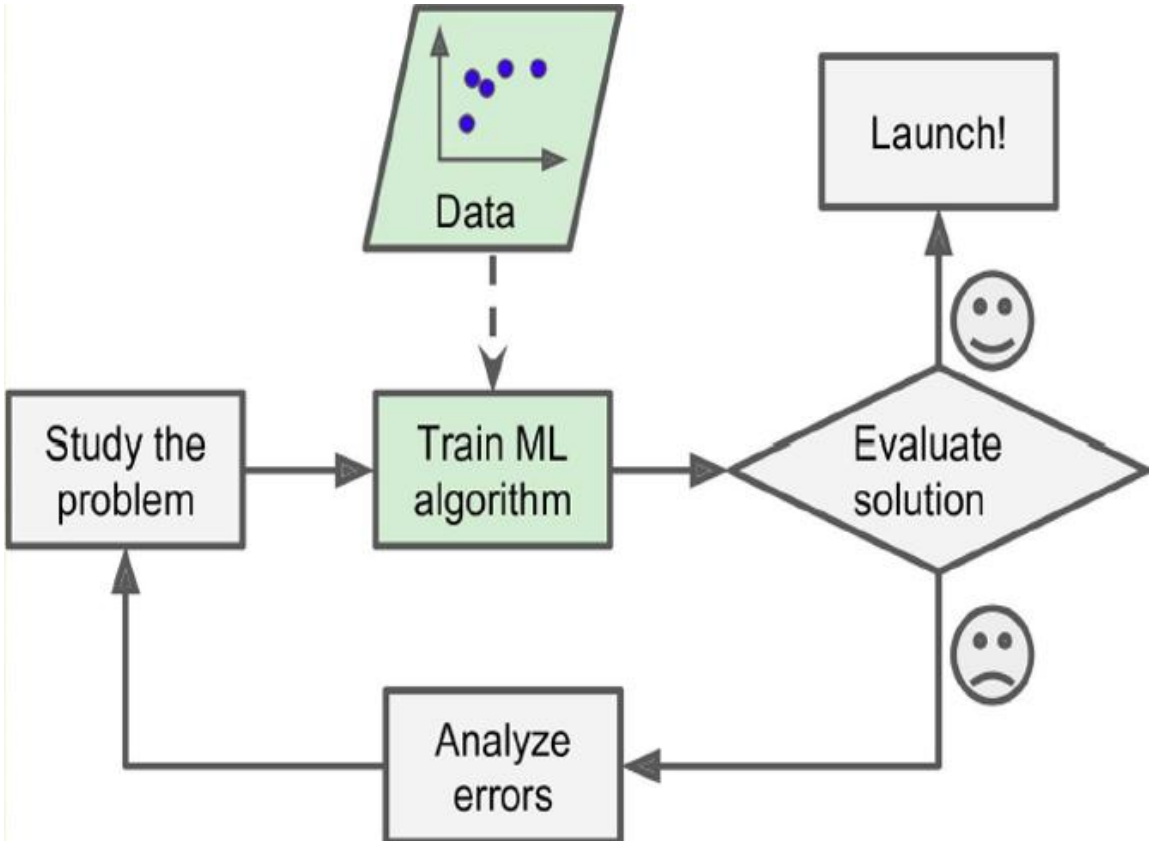
- Game Playing
- Robotics
- Self-Driving Cars
- Recommendation Systems



Machine Learning-Types



Machine Learning - Batch Learning



Machine Learning - Batch Learning

- Offline learning

- System is incapable of learning incrementally
  - It must be trained offline using all the available data

- Takes lot of time and computing resources
  - everytime training happens on the entire data

Machine Learning - Batch Learning

- Once the system is trained, it gets
  - Pushed to production
  - Runs without learning anymore
  - Just applies what it has learned offline

How to train the system with new data?

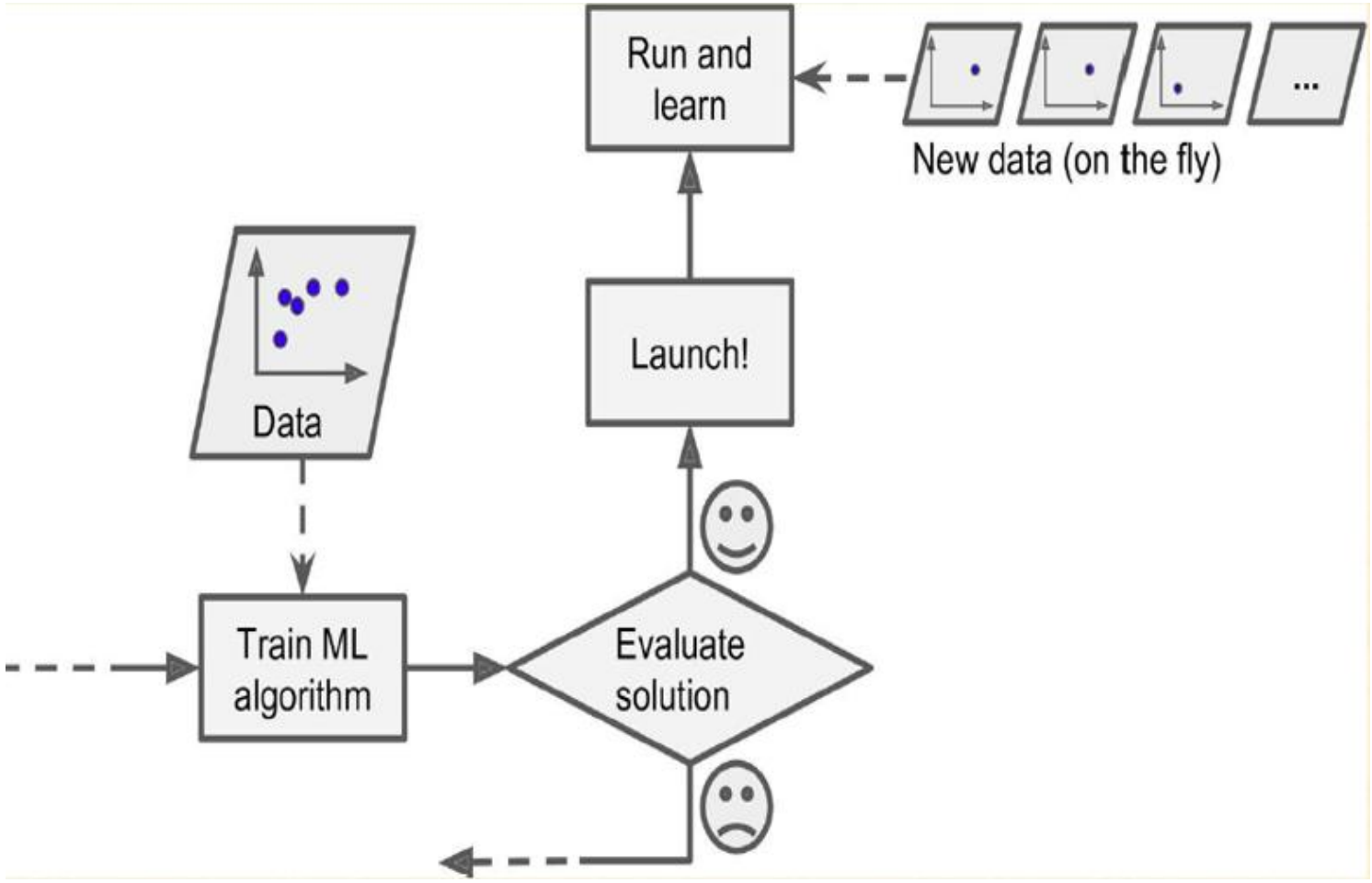
- We can not train the system incrementally
- We have to train the new system
  - From scratch on the full dataset
  - Stop the old system and
  - Replace it with the newly trained system

Machine Learning - Batch Learning

Limitations

- Since we have to train new system every time new data is available
  - Takes lot of computing resources
  - If the data is huge then
  - Training from scratch is nearly impossible
  - Becomes showstopper for lot of practical use cases

Machine Learning - Online Learning

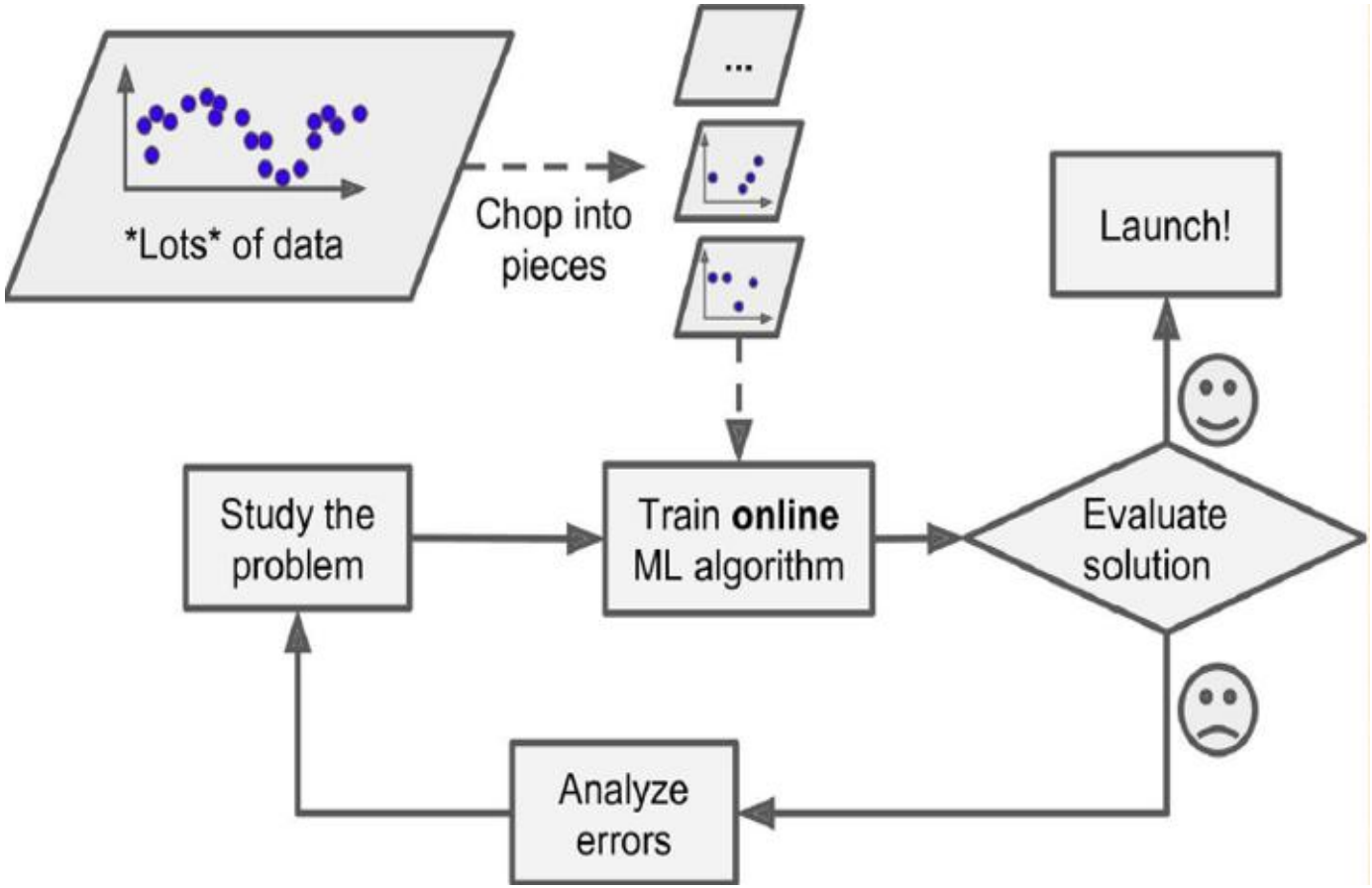


Machine Learning - Online Learning

- Train system incrementally
  - By feeding new data sequentially
  - Or in batches
- System can learn from new data on the fly
- Good for systems where data is a continuous flow
  - Stock prices

Machine Learning - Online Learning

Using online learning to handle huge datasets



Machine Learning - Online Learning

Using online learning to handle huge datasets

- Can be used to train huge datasets
  - That can not be fit in one machine
  - The training data gets divided into batches and
  - System gets trained on each batch incrementally

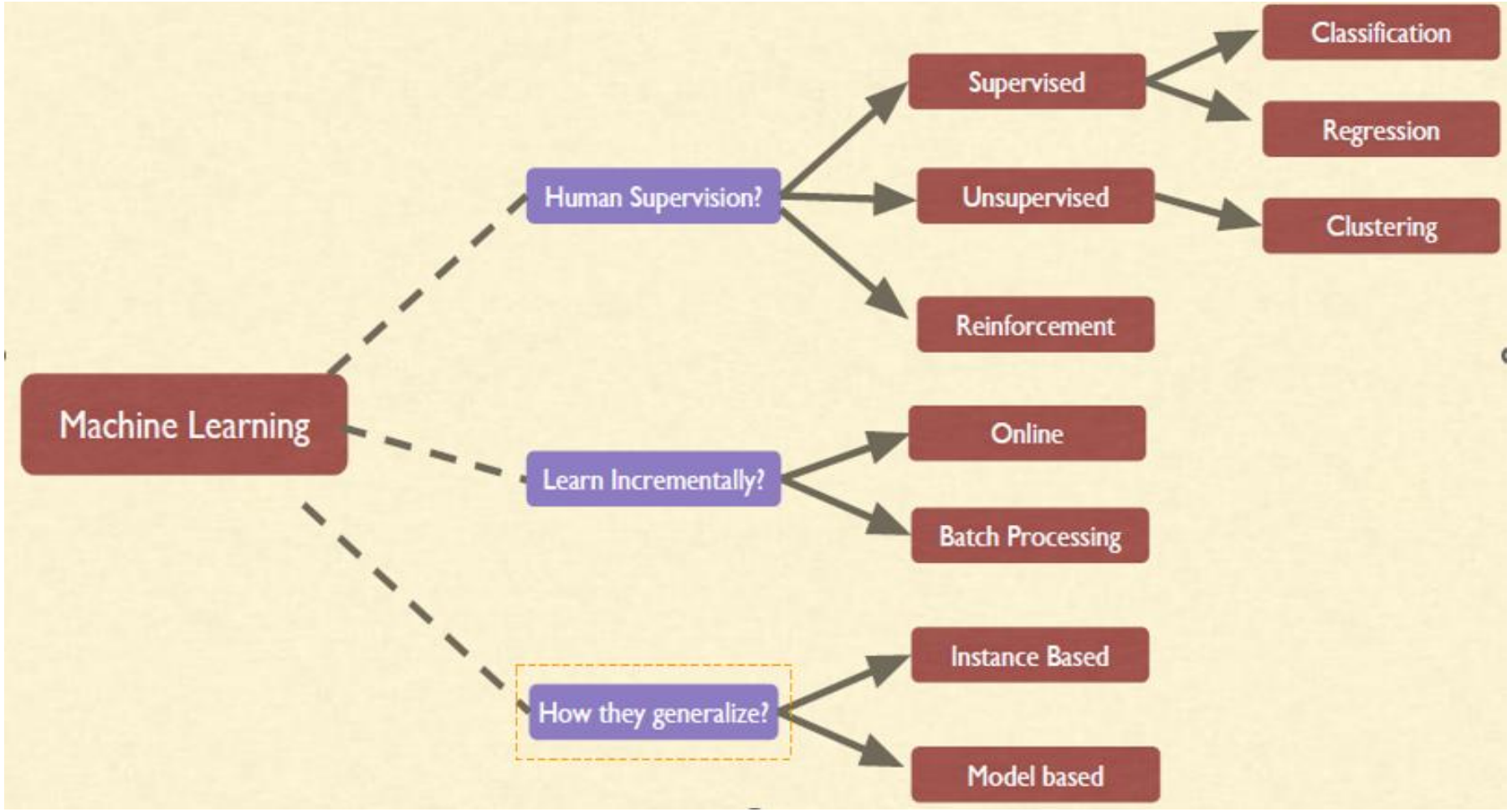
Machine Learning - Online Learning

Challenges in online learning

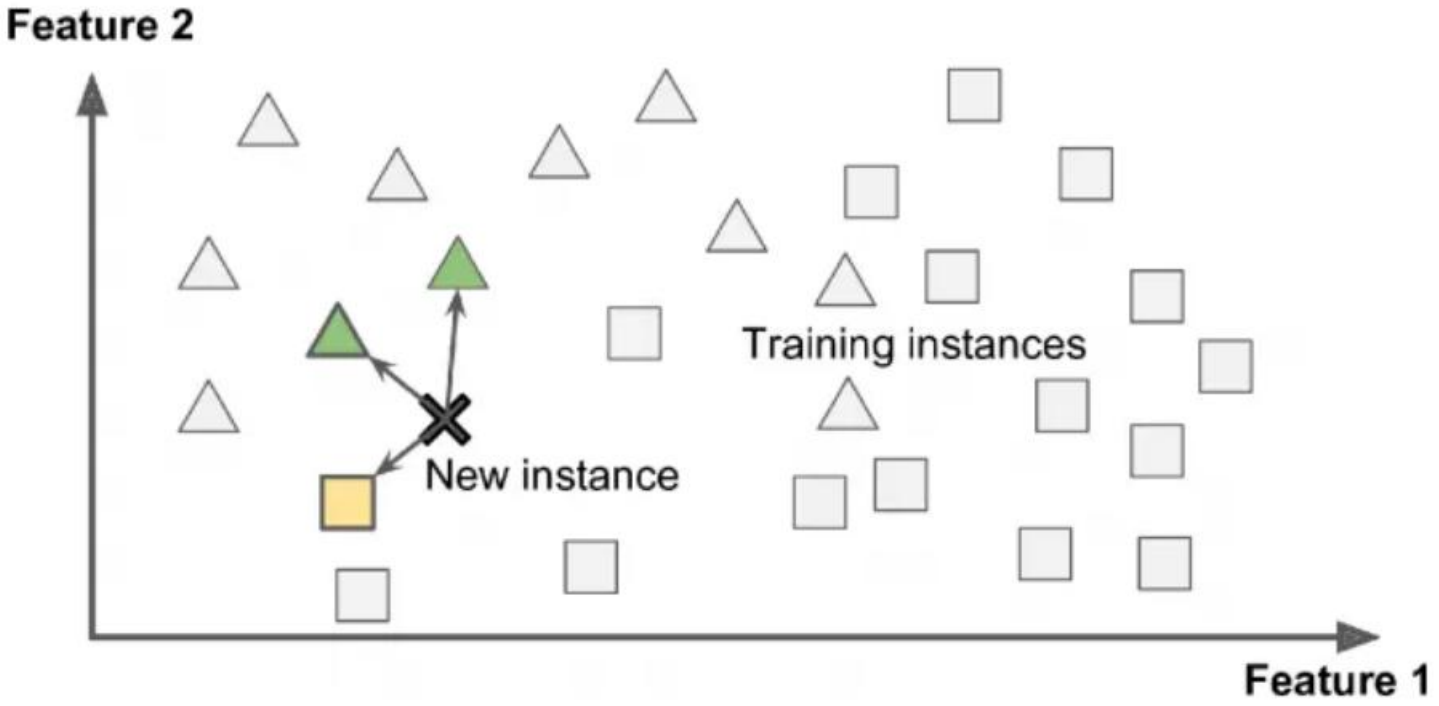
- System's performance gradually declines
  - If bad data is fed to the system
  - Bad data can come from
  - Malfunctioning sensor or robot
  - Someone spamming your system

- Closely monitor the system
  - Turn off the learning if there is a performance drop
  - Or monitor the input data and remove anomalies

Machine Learning-Types

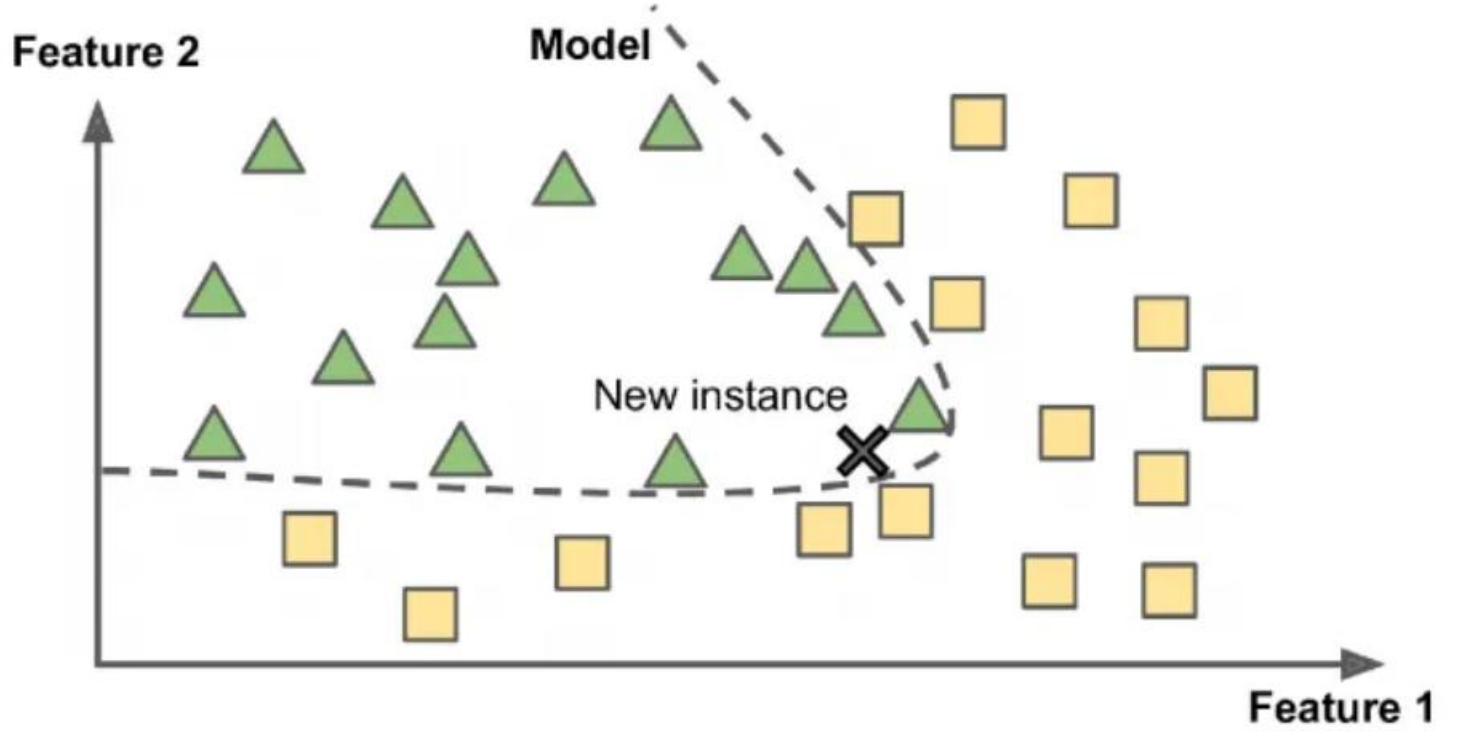


Machine Learning-Instance-Based Learning



The new instance would be classified as a triangle because the majority (2 Triangles vs 1 Square) of the most similar instances belong to that class.

Machine Learning-Model-based learning

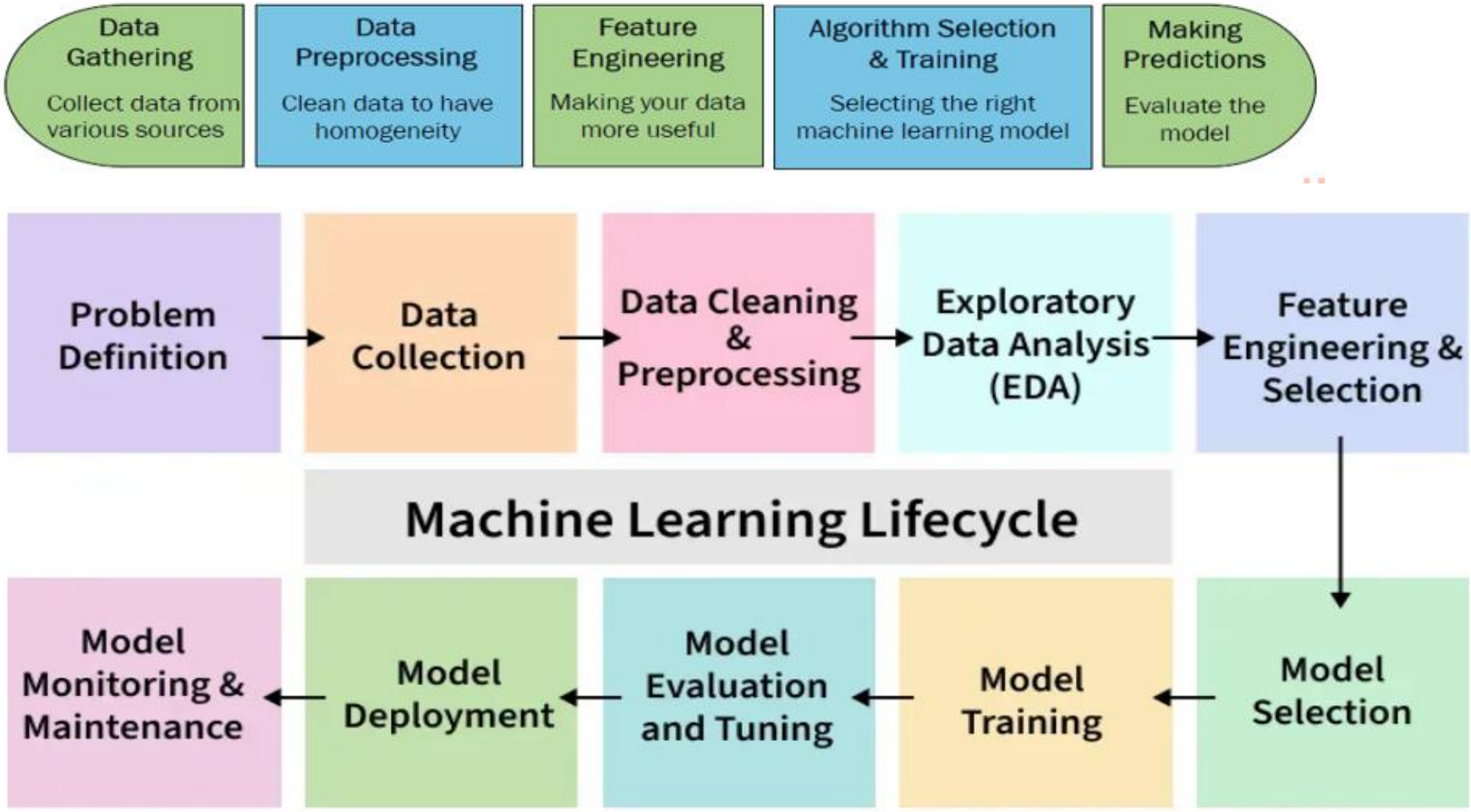


The new instance would be classified as a triangle no matter the majority of the most similar instances belong to Squar(1 Triangle vs 2 Squares) as the model defines so.

Model-based learning vs Instance-Based Learning

Model-based learning	Instance-Based Learning
Train model from training data to estimate model parameters i.e. discover patterns	Do not train model. Pattern discovery postponed until scoring query received
Store the model in suitable form	There is no model to store
Predict for unseen scoring instance using model	Predict for unseen scoring instance using Training directly
Can throw away input/training data after model training	Input/training data must be kept since each query uses part or full set of training observations
Storing models generally requires less storage	Storing training data generally requires more storage

Steps to Solve Machine Learning Problem



Machine Learning Lifecycle

THANK YOU